Dual-System Cognitive Model Flow (`apps/ai-service`)

Project: FLARE-CHRONOS — *Physics-Guided Cross-Sensor Harmonization of MODIS & VIIRS Hot Spots (2000–2026)*
Scope: End-to-end architecture, execution flow, deterministic tool gates, human-in-the-loop sign-off, and empirical evaluation of the Dual-System Cognitive AI Harness (`System 1 Reflexive Layer` + `System 2 Deliberative Layer`).
Version: `v1.1.0` — Validated on `15/15` Use Cases (`5` System 1 + `10` System 2) across both offline deterministic (`0` network calls) and live hosted-model (`Gemini 3 Flash`) execution modes.

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1. Architectural Purpose & Core Invariants

The cross-sensor physical harmonization engine (`packages/harmonization-core`) solves the satellite physics: spherical scan-angle footprint geometry, Solar Zenith Angle ($\mathbf{SZA} < 85^\circ$) unification, South Atlantic Anomaly (**SAA**) cosmic-ray filtering, two-stage Fire Radiative Power (**FRP**) calibration, $\tau = \mathbf{60\ min}$ multi-satellite episode clustering, and observation effort normalization $\mathbf{\Omega(y, \varphi)}$.

On top of that deterministic physics core, operational agencies (forest rangers, wildfire incident commanders, ministries of forestry, and climate scientists) ask natural-language situational and causal questions. A standard free-form Large Language Model (LLM) is unsafe for wildfire decision support because it can hallucinate hotspot counts, fabricate FRP values, or recommend unverified field dispatches.

The **FLARE-CHRONOS Dual-System Cognitive AI Harness (`apps/ai-service`)** eliminates LLM arithmetic and factual hallucination through **six hard architectural invariants**:

1. **Zero LLM Arithmetic (`Deterministic Tool Gate`)**: The LLM is strictly forbidden from computing physical, geometric, radiometric, or statistical quantities. Every number is computed by `18` registered deterministic `@tool` functions (`9` Atomic Tools + `9` Domain Diagnostic Tools, plus `1` internal raw-CSV harmonization tool = `19` `@tool` functions in total) backed by `flare_chronos_core`.
2. **Calibrated Reflexive Triage (`System 1 Reflexive Layer`)**: System 1 answers five closed-world classification questions (*region*, *epoch year*, *stakeholder agency*, *operational urgency*, and *Use Case assignment* among `15` canonical analyses) without generating prose. It runs a resilient three-step classifier chain:
    - **Primary (`typed_questions` decision endpoint)**: Returns calibrated token-level `logprob` probabilities.
    - **Fallback 1 (`openai_structured` / `native_chat` endpoint)**: Invokes forced structured tool-calling (`classify_request`) and records `triage_confidence_source = "self_reported"` so self-reported confidence is never mistaken for calibrated probability.
    - **Fallback 2 (`deterministic` extraction)**: Regex and keyword entity extraction so a run never fails even if all external model endpoints are unreachable.
3. **Zero-LLM Fast Reporting Tier (`System 1 Fast Execution`)**: Once a request is assigned to one of the `5` System 1 Use Cases (`UC-S1.1` through `UC-S1.5`) with calibrated confidence $\geq \mathbf{0.70}$, the `system1_fast_executor` invokes the single mapped Atomic Tool directly in $\sim \mathbf{1–5\ ms}$ with **zero LLM token generation** for prose.
4. **Model-Driven Deliberative Tool Orchestration (`System 2 Deliberative Layer`)**: For the `10` System 2 Use Cases (`UC-S2.1` through `UC-S2.10`)—or whenever System 1 confidence falls below **0.70** (`Dual-Process Symbiosis Escalation`)—the System 2 ReAct planner autonomously selects and chains tools across the full `18`-tool surface (`9` Atomic + `9` Domain Diagnostic).
5. **Verified Facts Ledger & Numeric Grounding Verifier**: Every scalar returned by a tool is recorded in the `VerifiedFactsLedger` with its `fact_id`, `tool_name`, `json_path`, `value`, `unit`, and `provenance_tag`. The model writes narrative prose using `{{cite:FACT_ID}}` placeholders; the harness deterministically injects the verified

numbers and rejects any draft containing an unverified number (`NumericGroundingVerifier`), forcing a self-correction loop or deterministic fallback.

6. **Identity-Bound Human-in-the-Loop Sign-Off (`human_sign_off_gate`)**: Any Use Case that proposes an operational field dispatch or policy sign-off (`UC-S1.4`, `UC-S2.6`, `UC-S2.9`, `UC-S2.10`) pauses the LangGraph state machine (`run_paused`) and emits a `sign_off_required` Server-Sent Event (SSE). The briefing is released (`Published`) only when an authenticated `Sign-off Principal` (extracted from the validated Bearer JWT, never from the request body) approves it; if rejected, the briefing is permanently `Withheld` (`final_briefing = ""`), and the decision is written to an append-only SQLite `audit` table protected by database triggers against `UPDATE` and `DELETE`.

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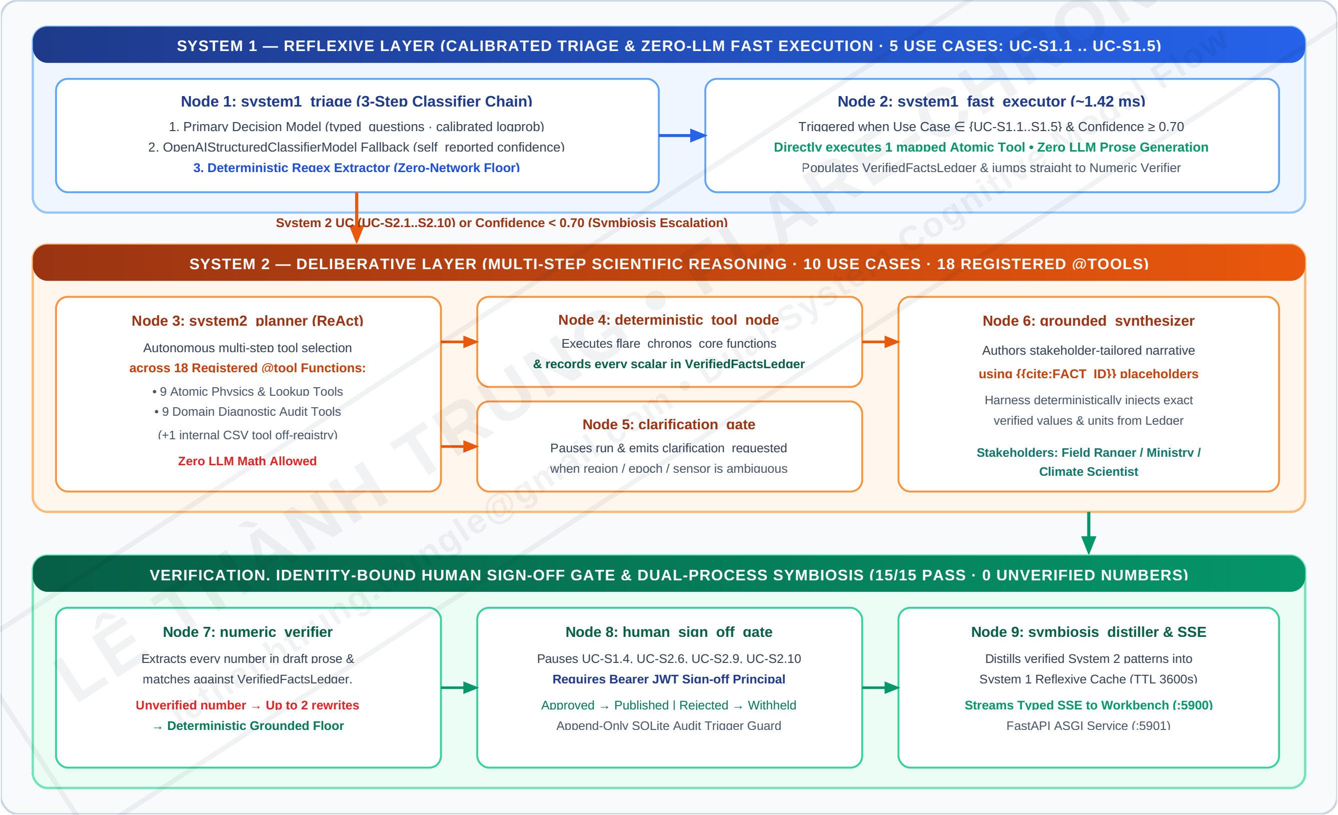
2. 8-Node LangGraph Dual-System StateGraph

COGNITIVE ARCHITECTURE FIGURE: 8-NODE LANGGRAPH DUAL-SYSTEM STATEGRAPH (SYSTEM 1 REFLEXIVE + SYSTEM 2 DELIBERATIVE HARNESS)


SYSTEM 1 — REFLEXIVE LAYER (CALIBRATED TRIAGE & ZERO-LLM FAST EXECUTION · 5 USE CASES: UC-S1.1 .. UC-S1.5)
Node 1: system1_triage (3-Step Classifier Chain)
1. Primary Decision Model (typed_questions · calibrated logprob)
2. OpenAIStructuredClassifierModel Fallback (self_reported confidence)
3. Deterministic Regex Extractor (Zero-Network Floor)
Node 2: system1_fast_executor (~1.42 ms)
Triggered when Use Case ∈ {UC-S1.1..S1.5} & Confidence ≥ 0.70
Directly executes 1 mapped Atomic Tool • Zero LLM Prose Generation
Populates VerifiedFactsLedger & jumps straight to Numeric Verifier
System 2 UC (UC-S2.1..S2.10) or Confidence < 0.70 (Symbiosis Escalation)
SYSTEM 2 — DELIBERATIVE LAYER (MULTI-STEP SCIENTIFIC REASONING · 10 USE CASES · 18 REGISTERED @TOOLS)
Node 3: system2_planner (ReAct)
Autonomous multi-step tool selection
across 18 Registered @tool Functions:
• 9 Atomic Physics & Lookup Tools
• 9 Domain Diagnostic Audit Tools
(+1 internal CSV tool off-registry)
Zero LLM Math Allowed
Node 4: deterministic_tool_node
Executes flare_chronos_core functions
& records every scalar in VerifiedFactsLedger
Node 5: clarification_gate
Pauses run & emits clarification_requested
when region / epoch / sensor is ambiguous
Node 6: grounded_synthesizer
Authors stakeholder-tailored narrative
using {{cite:FACT_ID}} placeholders
Harness deterministically injects exact
verified values & units from Ledger
Stakeholders: Field Ranger / Ministry /
Climate Scientist
VERIFICATION. IDENTITY-BOUND HUMAN SIGN-OFF GATE & DUAL-PROCESS SYMBIOSIS (15/15 PASS · 0 UNVERIFIED NUMBERS)
Node 7: numeric_verifier
Extracts every number in draft prose &
matches against VerifiedFactsLedger.
Unverified number → Up to 2 rewrites
→ Deterministic Grounded Floor
Node 8: human_sign_off_gate
Pauses UC-S1.4, UC-S2.6, UC-S2.9, UC-S2.10
Requires Bearer JWT Sign-off Principal
Approved → Published | Rejected → Withheld
Append-Only SQLite Audit Trigger Guard
Node 9: symbiosis_distiller & SSE
Distills verified System 2 patterns into
System 1 Reflexive Cache (TTL 3600s)
Streams Typed SSE to Workbench (:5900)
FastAPI ASGI Service (:5901)

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3. Canonical Taxonomy: 15 Use Cases & 18 Registered `@tool` Functions

Per ADR-0016, the cognitive harness is scoped to **15 non-overlapping Use Cases** (`5` System 1 Reflexive + `10` System 2 Deliberative) and **18 model-facing `@tool` functions** (`9` Atomic + `9` Domain Diagnostic), plus `1` internal raw-CSV harmonization tool (`tool_harmonize_real_firms_csv_slice`) kept outside the LLM registry so the language model never touches raw unharmonized CSV rows directly.

3.1. System 1 Reflexive Use Cases (`5` Analyses — `9` Atomic Tools Only)

Use Case ID	Analysis Name	Primary Atomic Tool	Sign-Off Gate	Target Stakeholder
`UC-S1.1`	**Intent Triage & Single-Epoch Lookup**	`tool_lookup_single_epoch_metrics`	No	Field Ranger / Analyst
`UC-S1.2`	**Spatio-Temporal Entity Extraction**	`tool_lookup_single_epoch_metrics`	No	Field Ranger / Analyst
`UC-S1.3`	**Single-Hop Historical Epoch Lookup**	`tool_lookup_single_epoch_metrics`	No	Forestry Manager
`UC-S1.4`	**Flash NRT Crisis Alert Triage**	`tool_emit_flash_nrt_alert`	**Yes**	Wildfire Incident Commander
`UC-S1.5`	**Stakeholder Briefing Card Synthesis**	`tool_assemble_stakeholder_briefing_card`	No	Climate Scientist

3.2. System 2 Deliberative Use Cases (`10` Analyses — Full `18` -Tool Surface)

Use Case ID	Analysis Name	Primary Domain Diagnostic Tool	Sign-Off Gate	Target Stakeholder
`UC-1.1`	**2024 Multi-Sensor Regime-Shift Attribution**	`tool_uc1_1_regime_shift_attribution`	No	Climate Scientist / Examiner
`UC-1.2`	**Cross-Sensor Scan-Angle Discrepancy Forensics**	`tool_uc1_2_cross_sensor_discrepancy_forensics`	No	Remote Sensing Scientist
`UC-1.3`	**Thermal Source Origin & SAA Noise Classification**	`tool_uc1_3_thermal_source_origin_classification`	No	Data Quality Engineer
`UC-2.1`	**Phenological Phase-Shift & SFAI Detection**	`tool_uc2_1_phenological_phase_shift_detection`	No	Climate Scientist
`UC-2.2`	**Uncertainty-Aware Tactical Field Dispatch**	`tool_uc2_2_uncertainty_aware_tactical_dispatch`	**Yes**	Wildfire Incident Commander
`UC-2.3`	**Verified Multi-Stakeholder Policy Briefing**	`tool_assemble_stakeholder_briefing_card`	**Yes**	Ministry of Forestry Director
`UC-3.1`	**2026+ Constellation Retirement Continuity Simulation**	`tool_uc3_1_constellation_retirement_simulation`	No	Space Agency Program Lead
`UC-3.3`	**Autonomous Biome Radiometric Recalibration**	`tool_uc3_3_autonomous_biome_recalibration`	No	Remote Sensing Scientist

Use Case ID	Analysis Name	Primary Domain Diagnostic Tool	Sign-Off Gate	Target Stakeholder
`UC-4.1`	**Orbital Drift & Astronomical SZA Twilight Audit**	`tool_uc4_1_orbital_drift_and_sza_audit`	No	Orbital Physics Analyst
`UC-4.2`	**Near-Real-Time (NRT) vs. Standard (SP) Fidelity Audit**	`tool_uc4_2_nrt_vs_sp_fidelity_audit`	No	Operations Analyst

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4. Empirical Evaluation: Deterministic & Live Hosted-Model Benchmarks

Both benchmark artifacts are committed under `data/processed/` and verified by `tests/test_ai_service_harness.py`: - **Deterministic Offline Benchmark (**`data/processed/ai_harness_results.json`**)**: Runs all `15` Use Cases with zero network calls (`0.0000 USD` cloud spend). - **Live Hosted-Model Benchmark (**`data/processed/ai_harness_results_live.json`**)**: Runs all `15` Use Cases against live **Google Gemini (**`gemini-3-flash-preview`**)** via `apps/ai-service/src/ai_service/run_all_usecases.py --live`.

Evaluation Metric	Deterministic Harness (`ai_harness_results.json`)	Live Hosted Model (`ai_harness_results_live.json`)	Architectural Guarantee
Total Use Cases Evaluated	`15 / 15` (`100.0%`)	`15 / 15` (`100.0%`)	`5` System 1 + `10` System 2
Schema & Invariant Pass Rate	`15 / 15` (`100.0%`)	`15 / 15` (`100.0%`)	Strict Pydantic validation
Unverified Numeric Claims Released	`0`	`0`	Enforced by `NumericGroundingVerifier`
Unresolved Citation Placeholders	`0`	`0`	Verified against `VerifiedFactsLedger`
System 1 Mean Latency	`1.42 ms`	`2,718.6 ms` *(includes remote classification)*	Zero LLM prose generation in System 1
System 2 Mean Latency	`1.61 ms`	`11,264.8 ms` *(multi-round ReAct + synthesis)*	Autonomous tool chaining (`1.0` tool round avg)
Triage Confidence Source	`logprob` (`5/5` S1, `10/10` S2)	`self_reported` (`15/15` via Gemini fallback)	Honest provenance when local `:11434` is offline
Classifier Chain Resilience	`0` fallbacks	`15/15` graceful fallbacks from `:11434` to Gemini	Run never fails when local decision node is down
Total Token Consumption (`15` UCs)	`0` tokens	`31,910` prompt + `2,473` completion = **`34,383` tokens**	Grounded synthesis only in System 2

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⚡ TECHNICAL & ARCHITECTURAL NOTE

Phiên bản: `v1.1.1` — cập nhật 04/11/2026 **Transparency on Classifier Telemetry in `ai_harness_results_live.json`:** During the live benchmark run, the optional local `SystemOneChatModel` tunnel (`http://127.0.0.1:11434/v1/systemone`) was intentionally offline. As designed, all `15` runs automatically recorded `triage_fallback: "SystemOneChatModel_unavailable"` and fell back to the secondary `OpenAIStructuredClassifierModel` (`gemini-3-flash-preview`), achieving `15/15` **accurate Use Case assignments** while stamping `triage_confidence_source: "self_reported"` so downstream reviewers know the confidence came from the chat fallback rather than token logprobs.

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5. Reproducing the Model Flow & Live SSE Stream

5.1. Start the FastAPI Analysis Server (`:5901`) and Next.js Workbench (`:5900`)

```
# Terminal 1 — Start the FastAPI ASGI service on port 5901 (local demo mode)
AI_SERVICE_AUTH=disabled uv run python -m ai_service.serve

# Terminal 2 — Start the interactive Next.js 16 Workbench on port 5900
cd apps/workbench && bun run dev
```

5.2. Stream a Live Run & Inspect Human Sign-Off via `curl`

```
# 1. Stream a System 1 crisis flash alert (pauses at human_sign_off_gate)
curl -N "http://127.0.0.1:5901/api/v1/analysis/stream?use_case_id=UC-S1.4&query=UC-S1.4+flash+alert+for+amazon+2026&thread_id=demo-signoff"

# 2. Resume the paused run with an identity-bound approval decision
curl -X POST "http://127.0.0.1:5901/api/v1/analysis/resume" \
  -H "Content-Type: application/json" \
  -d '{"thread_id": "demo-signoff", "approved": true, "reason": "Verified against Pentad 47 SFAI baseline"}'
```

5.3. Run Offline & Live Test Suites

```
# Offline test suite (182 tests across 22 files; 0 network calls)
uv run python -m pytest -q

# Live hosted-model vertical slice suite (6 live TDD seam tests)
uv run python -m pytest -m live -q
```